EXPERIMENT – 5

Performing Various Configurations of IPTables Firewall in Kali Linux

Aim:

To perform and analyze various firewall configurations using IPTables in Kali Linux to control network traffic and enhance system security.

Software Requirements:

- **Host Operating System:** Windows / Linux / macOS
- **Virtualization Software:** Oracle VirtualBox
- **Guest Operating System:** Kali Linux
- **Firewall Tool:** IPTables
- **System Requirements:**
    - RAM: Minimum 4 GB (8 GB recommended)
    - Disk Space: Minimum 30 GB
    - Processor: Intel/AMD with Virtualization support

Procedure:

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

Step 2: Open Terminal and Check IPTables

1. Open Terminal in Kali Linux.
2. Check IPTables version:
   iptables --version

Step 3: View Existing Firewall Rules

1. Display current IPTables rules:
   sudo iptables -L

Step 4: Set Default Policies

1. Set default policy to drop incoming traffic:
   sudo iptables -P INPUT DROP
2. Allow outgoing traffic:
   sudo iptables -P OUTPUT ACCEPT

Step 5: Allow Localhost Traffic

1. Allow loopback interface traffic:
   sudo iptables -A INPUT -i lo -j ACCEPT

Step 6: Allow SSH Access

1. Allow SSH connections on port 22:
   sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT

Step 7: Allow HTTP and HTTPS Traffic

1. Allow HTTP traffic:
   sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
2. Allow HTTPS traffic:
   sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT

Step 8: Block a Specific IP Address

1. Block traffic from a specific IP:
   sudo iptables -A INPUT -s 192.168.1.100 -j DROP

Step 9: Save IPTables Rules

1. Save the configured rules:
   sudo iptables-save > firewall_rules.txt

Output:

```
┌──(root㉿kali)-[~]
└─# sudo iptables -A INPUT -s 192.168.1.100 -j DROP

┌──(root㉿kali)-[~]
└─# sudo iptables -L INPUT -n | grep DROP
Chain INPUT (policy DROP)
DROP       all  --  192.168.1.100        0.0.0.0/0
```

```
└─# cat firewall_rules.txt
# Generated by iptables-save v1.8.11 (nf_tables) on Wed Apr 22 13:09:07 2026
*filter
:INPUT DROP [0:0]
:FORWARD ACCEPT [0:0]
:OUTPUT ACCEPT [0:0]
-A INPUT -i lo -j ACCEPT
-A INPUT -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
-A INPUT -p tcp -m tcp --dport 22 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 80 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 443 -j ACCEPT
-A INPUT -s 192.168.1.100/32 -j DROP
COMMIT
# Completed on Wed Apr 22 13:09:07 2026
```

Result:

Various firewall configurations were successfully implemented using IPTables in Kali Linux.